Chun **Zhou**

Shenzhen Foreign Languages School, Shenzhen, China

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Education

Shenzhen Foreign Languages School — Shenzhen, China

High School Education — Sep. 2024 – Present

- GPA: 4.18/4.30.
- **9 (10) AP scores of 5:** Calculus BC (AB subscore), Statistics, Physics C: Electricity and Magnetism, Physics C: Mechanics, Physics 1, Physics 2, Chemistry, Macroeconomics, and Microeconomics.
- **3 AP scores of 4:** Computer Science A, AP Seminar, and World History: Modern. **SAT:** 1530.

Shenzhen Nanshan Experimental Education Group Qilin Middle School — Shenzhen, China

Middle School Education — Sep. 2021 – Jul. 2024

- Average score for Grade 9: 93/100.

Honors & Awards

Grade 11	**Silver, Top 2–3**, Rhino-Bird Science Talent Advancement Program	International
Grade 11	**Commendation (shortlisted)**, John Locke Institute 2026 Global Essay Prize, Law Category	International
Grade 11	**Nominee**, TechExplorer Camp in Shanghai, SpringSense Innovation Institute — A highly selective and fully sponsored winter camp for young tech students teaching computer science	National
Grade 10	**National Gold, Division 1**, 39th AAPT Physics Bowl Contest	International
Grade 11	**Global Silver**, British Physics Olympiad, Round 1	International
Grade 10	**Distinction (Top 5%) and AIME Qualifier**, AMC 12	International
Grade 10	**Gold Award**, BPhO Online Senior Physics Challenge	International

Research Experience

Learning Conjugate Fringe Decomposition for Single-shot Fringe Projection Profilometry — Rhino-bird Program, Tencent-sponsored

Independent Researcher — Grades 11–12

- Conducted a research program through the Rhino-Bird Science Talent Advancement Program; proposed a novel physics-informed neural network (PINN) for 3D surface measurement.
- Sole author of a manuscript accepted by **Applied Optics, a JCR Q3 peer-reviewed journal indexed in the Science Citation Index**. DOI: 10.1364/AO.611321. Independently completed the entire research and publication process, from developing the research question and conducting experiments to writing the manuscript, submitting it, and responding to reviewers.

Learning Conjugate Compensation for Neural OFDM Signal Receiver — Independent Research

Independent Researcher — Grade 12

- Inspired by the first PINN research experience, conceived a robust architecture for a neural OFDM signal receiver to improve accuracy under in-phase/quadrature imbalance.
- Prepared a synthetic dataset and tested the architecture across three different backbones. Wrote a short research letter summarizing the idea; it is under consideration at **IEEE Signal Processing Letters, a CCF-C-tier letter journal**.

Deep Learning Method for Recovering a Time Sequence from a Single Frame — Cooperative Research

Student Researcher & Co-author — Grades 10–11

- Verified a temporal super-resolution mechanism based on structured-light projection and frequency-domain multiplexing. Under supervision, reproduced both a neural network and conventional demodulation methods.

Technical Projects & Open Source

MATLAB Central File Exchange: Seven Published Tools — Online

Independent Open-source Developer — Grades 10–12

- Fourier 3D Display; Picasso Style Image Maker; Cyberpunk Style Image Maker; Audio Denoising Processor; Snowy Fog Image Processor; Monte Carlo Pi Estimator; Magnetic Newton's Cradle Simulator.

GitHub Repository: A COMSOL Model of the Annular Well Problem — Online

Contributor: GitHub repository — Grade 11

- A COMSOL-simulated model based on Euler–Euler two-phase flow. It can be used to study optimization of annular ratio and initial velocity; inclination, cuttings, liquid properties, and geometry are modeled.

Skills

Programming Python, MATLAB, COMSOL | **Compile:** LaTeX, Office Suite, OriginLab, draw.io